

Ly Chungheang

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5 November 2004 (21 years old)

[My Portfolio](#) |  [Chungheang0980](#)

N.109, Orkede Village, Sangkat Ou Bek K'am, Khan Sen Sok,
Phnom Penh, Cambodia



SUMMARY

I'm currently 21 years old. Motivated student at the Institute Technology of Cambodia 4th year in major Data Science with a growing foundation in data analysis and administrative operations. Skilled in managing tasks efficiently, maintaining confidentiality, and fostering effective communication. Strong organizational abilities and a commitment to learning enable seamless coordination of projects and office workflows. Eager to apply academic knowledge to real-world data challenges and contribute to team success.

EDUCATION

Institute of Technology of Cambodia
Engineer, Data Science

Expected Graduation: August 2027
Phnom Penh, Cambodia

Institute of foreign languages
Adults English Program (AEP)

Graduated: March 2026
Phnom Penh, Cambodia

Lycee Preah Angduong
High School Diploma

Graduated: 2022
Prey Veng Province

EXPERIENCE AND VOLUNTEER

Khmer OCR and Plate Recognition
ReDA Laboratory, ITC

June 2025 – August 2025
Phnom Penh, Cambodia

- Developed a Khmer license plate detection and OCR system for cars and motorbikes using computer vision and OCR techniques.
- Collected and prepared image data in different real-world conditions, including daylight, night, rain, and dusty environments.
- Applied YOLOv8 for license plate detection and CRNN-Keras for recognizing Khmer and Latin characters on plates.
- Planned the full system pipeline from image/video input to plate detection, OCR recognition, and deployment through Streamlit/Flask or embedded devices.

22nd CamTESOL (IT Support)
Institute of Technology of Cambodia, Building I

8 February 2025– 10 February 2025
Phnom Penh, Cambodia

- Supported CamTESOL 2025 Conference in Cambodia as an IT support staff member.
- technical setup for conference sessions, including computers, projectors, and audio equipment.
- Helped speakers and participants resolve basic technical issues during the event.
- Monitored IT equipment to ensure smooth conference operations.

BACLL Organizer-ACU

Lycee Preah Angduong

28 August 2025– 29 August 2025

Prey Veng, Cambodia

- Assisted with participant registration, communication, and on-site event support.
- Helped manage event logistics to ensure sessions ran on time and participants had a positive experience.

Teaching Coordinator (Volunteer)

The Church of Jesus Christ of Latter-day, TSK

February 2024 – March 2024

Phnom Penh, Cambodia

- Taught self-reliance classes at TSK Church to help members improve personal growth and responsibility.
- Prepared and shared lessons on goal setting, planning, and independent problem-solving.
- Supported learners in building confidence, discipline, and self-management skills.

PROJECTS

Unemployment Prediction Using Economic Indicators

May 2026 – Now

Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook

- Developed a machine learning model to predict unemployment rates using economic and labor market data from World Bank, FRED, and ILO datasets covering 200+ countries.
- Engineered time-series features including lagged unemployment rates, rolling averages, GDP indicators, and labor force metrics to improve prediction accuracy.
- Trained and compared multiple regression and ensemble models, achieving an R^2 score of 0.976 with the best-performing Linear Regression model.
- Built interactive visualizations and analytical dashboards to evaluate model performance, feature importance, and global unemployment trends.
- Present research findings at an academic conference in Malaysia and published a paper on unemployment prediction and economic analytics.
- Source code: [Unemployment Economic Indicators](#) 

Real-Time Stock Market Data Pipeline Forecasting Dashboard

May 2026 – June 2026

Python, PySpark, PostgreSQL, FastAPI, Streamlit, MLflow, Docker

- Built a real-time stock market analytics using Python, PySpark, PostgreSQL, FastAPI, and Streamlit.
- Developed an end-to-end ETL pipeline to collect, transform, and store stock market data from the Finnhub API.
- Designed a multi-layer data warehouse (Bronze, Silver, Gold) for scalable analytics and reporting.
- Implemented machine learning forecasting models (Linear Regression and Holt's Smoothing) with MLflow experiment tracking.
- Created interactive dashboards and REST APIs for real-time monitoring, historical analysis, and stock price prediction.
- Source code: [Unemployment Economic Indicators](#) 

SecureBG Banking Web System

April 2026 – May 2026

Python, Flask, SQLite, Docker, XML-RPC, HTML/CSS, JavaScript

- Developed a secure banking web application using Flask, SQLite, and Docker.
- Implemented encrypted authentication, password hashing, and role-based access control.
- Built banking features including deposits, withdrawals, transfers, and transaction history.
- Integrated fraud detection and audit logging to monitor suspicious activities.
- Demo: [Banking Security](#) 

Bank Transaction Scanner

March 2025 – June 2025

Python, FastAPI, Next.js, TensorFlow, Tesseract OCR, MongoDB, Docker

- Developed an OCR-based system to extract transaction details from bank receipt images automatically.
- Built a CNN model to classify ABA and ACLEDA receipts before data extraction.
- Implemented automated extraction of transaction ID, date, amount, and currency using OCR and validation pipelines.
- Developed a full-stack web application using FastAPI, Next.js, and MongoDB for receipt processing and report export.
- Successfully presented the project at the 14th Scientific Day Showcase and was selected as one of the outstanding candidates.
- Source code: [Bank-Transaction Scanner](#) 
- Video Demo and Poster: [Demo](#), [Poster](#)

SKILLS

Programming: Python, JavaScript, SQL, $\text{\LaTeX}$

ML/AI: PyTorch, TensorFlow, scikit-learn, TensorFlow, Scikit-learn, YOLOv8, CNN, Tesseract OCR

Data Analysts: Excel, Matplotlib, R-Programming, PowerBI

Web & Systems: React, Django, HTML/CSS, Node.js, NestJS, PostgreSQL, Docker

Specialized Area: Statistics, Machine Learning, Data Scientist, Machine Learning, Computer Vision

Tools & Practices: Git, GitHub, Jupyter Notebook, MLflow

Languages: English (*Upper intermediate*), Khmer (*native*), French (*Elementary*)

CERTIFICATIONS

- [Data Privacy Fundamental](#)
- [Python 101 for Data Science](#)
- [14th Scientific Day](#)
- [BACLL Organizer-ACU](#)
- [22nd CamTESOL Conference](#)

REFERENCES

1. **Dr. LONG Pakrigna**
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Institute of Technology of Cambodia
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